C9555 A Survival Guide to Simulations

Free Energy Calculations, Part 1

Ivo Kabelka

PLUMED Introduction

- A powerful open-source library
- Provides tools for:
  - Enhanced sampling and free energy calculations
  - Trajectory analysis
- Not tied to a specific MD engine



PLUMED

The community-developed PLUgin for MolEcular Dynamics

https://www.plumed.org/

MUNI | CEITEC

Using PLUMED

- PLUMED can be used with GROMACS
  - Significantly extends functionality
  - Negatively impacts performance
- GROMACS needs to be patched first
  - PLUMED is compiled
  - PLUMED is installed
  - GROMACS is patched with PLUMED patch tool

```bash
$ plumed patch
```

  - Patched GROMACS is installed

```bash
$ gmx mdrun -deffnm md -plumed plumed.dat
```


MUNI | CEITEC

Using PLUMED

- PLUMED can be used as a standalone tool
  - Irrespective which MD engine was used to produce the trajectory
  - Several command line tools, such as **driver** or sum_hills
- To analyze a simulation snapshot in GRO format

```bash
$ plumed driver --plumed plumed.dat --igro snapshot.gro
```

- To analyze a trajectory in XTC format

```bash
$ plumed driver --plumed plumed.dat --mf_xtc traj.xtc
```


MUNI | CEITEC

PLUMED Input File Format

- Structured and executed in an action-based format
  - Each action can consist of multiple tasks or routines that are not visible to the user

- Actions are referenced by their labels, such as dist
  - Some actions also have components, such as dist.x
  - Components are accessed using a dot notation

PLUMED Input File Format

```
# This is a comment
# Definitions of index groups
sys: GROUP NDX_FILE=index.ndx NDX_GROUP=System UNIQUE
ion: GROUP NDX_FILE=index.ndx NDX_GROUP=Ion UNIQUE

# Creates a virtual atom in the center of the simulation box
c_sys: CENTER ATOMS=sys

# Writes a file with coordinates
DUMPATOMS FILE=sys.gro ATOMS=c_sys

# Calculates the distance between the virtual atom and an ion
dist: DISTANCE ATOMS=c_sys,ion COMPONENTS

PRINT ARG=dist.x FILE=DIST
FLUSH STRIDE=100
```

dat (plumed)

MUNI | CEITEC

PLUMED Input File Format

- PLUMED does not directly work with topology information

```
MOLINFO STRUCTURE=system.pdb
```
dat (plumed)

- Atoms can be references using indices
  - Numbering is consistent with most MD engines (1-based)
  - Inconsistent with VMD (0-based)

```
sys: GROUP ATOMS=1,2,3,4,5, …
```
dat (plumed)

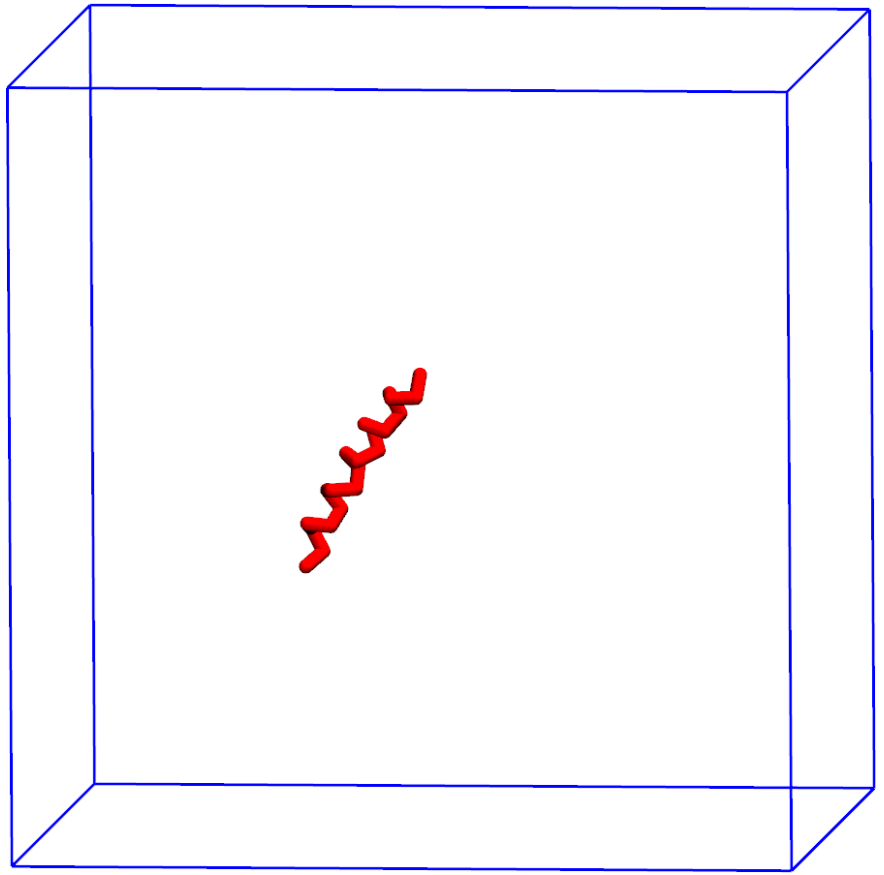
```
sys: GROUP NDX_FILE=index.ndx NDX_GROUP=System UNIQUE
```
dat (plumed)

  - UNIQUE removes duplicate indexes, which can cause errors or calculation overhead

MUNI | CEITEC

The Toy System

Calculating the position of the peptide



MUNI | CEITEC

PLUMED Input File Format

- Calculate the distance between the peptide and box center
  - Only along the X-axis

```
# Indexes of all beads in the system are read from the ndx file
sys: GROUP NDX_FILE=index.ndx NDX_GROUP=System UNIQUE

# Creates a virtual atom in the center of the simulation box
c_sys: CENTER ATOMS=sys

# Calculates the distance between the system center and an atom
dist: DISTANCE ATOMS=c_sys,11 COMPONENTS

PRINT ARG=dist.x FILE=DIST
FLUSH STRIDE=100
```

dat (plumed)

MUNI | CEITEC

Generating Index File Using GROMACS

```bash
$ gmx make_ndx -f equi.gro -o index.ndx << EOF
del 1-13
a BB
name 1 Peptide
ri 1
name 2 NTerm
ri 11
name 3 Pep_CoM
ri 21
name 4 CTerm
a W
name 5 Water

q
EOF
```


MUNI | CEITEC

Generating Index File Using GROMACS

Automatically identified index groups

```
0 System          : 7881 atoms
1 Protein         : 21 atoms
2 Protein-H       : 21 atoms
3 C-alpha         : 0 atoms
4 Backbone        : 0 atoms
5 MainChain       : 0 atoms
6 MainChain+Cb.   : 0 atoms
7 MainChain+H     : 0 atoms
8 SideChain       : 21 atoms
9 SideChain-H.    : 21 atoms
10 Prot-Masses    : 21 atoms
11 non-Protein    : 7860 atoms
12 Other          : 7860 atoms
13 W              : 7860 atoms
```

Newly generated index groups

```
0 System      : 7881 atoms
1 Peptide     : 21 atoms
2 NTerm       : 1 atoms
3 Pep_CoM     : 1 atoms
4 CTerm       : 1 atoms
5 Water       : 7860 atoms
```


MUNI | CEITEC

Generating Index File Using GROMACS

- Each index group has a name enclosed in square brackets

```
[ Peptide ]
  1   2   3   4   5   6   7   8   9 10 11 12 13 14 15
16 17 18 19 20 21
[ NTerm ]
  1
[ CTerm ]
21
```

ndx (gromacs)

- PLUMED will not read group names not enclosed by spaces

MUNI | CEITEC

Setup for the Practical Exercises

```bash
# Local temporary data storage
$ cd /scratch/${USER}

# Make new commands available: gmx and plumed
$ module add gromacs:2021.4-plumed

# Make new command available: vmd
$ module add vmd

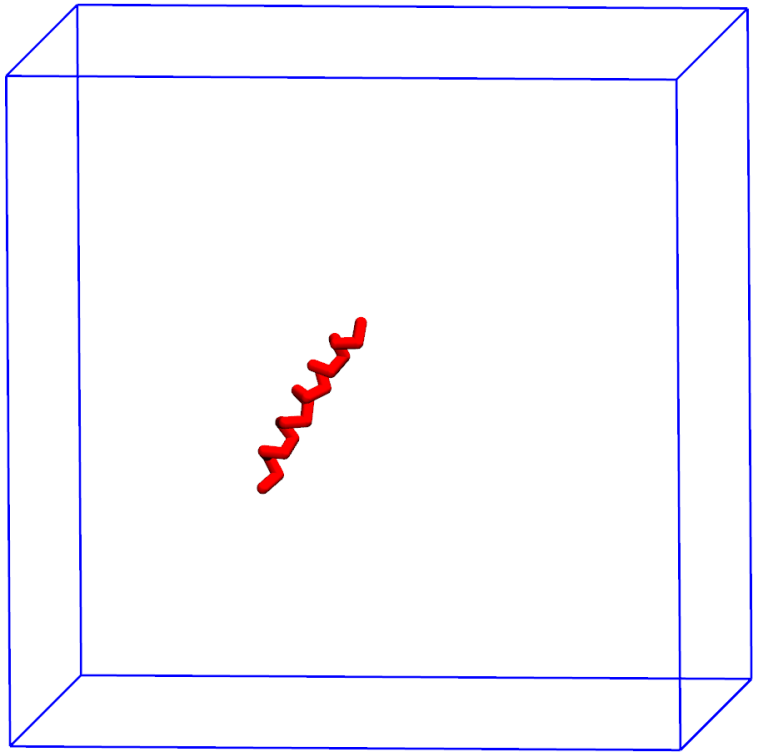
# Copy data for today's class
cp -r /storage/wolf/home/ivo/C9555/03 .
```


MUNI | CEITEC

Task 1

Peptide in water

- Calculate the distance between the box center and peptide atom



MUNI | CEITEC

Task 1

Peptide in water

- Calculate the distance between the box center and peptide atom
  - Generate index.ndx file using gmx make_ndx
  - Define atom groups in plumed

    Pep_CoM: GROUP ATOMS=[...]

    System: GROUP NDX_FILE=[...] NDX_GROUP=[...]
  - Calculate the distance along the x-axis
  - Write the results in a file

```
sys: GROUP [...]
c_sys: CENTER [...]
dist: DISTANCE [...]
PRINT ARG [...]
```

dat (plumed)

```
$ plumed driver --plumed plumed.dat --mf_xtc traj.xtc
```

bash

MUNI CEITEC

Calculating the Distance

```
# Calculates the distance between the system center and an atom
dist: DISTANCE ATOMS=c_sys,11 COMPONENTS

PRINT ARG=dist.x FILE=DIST
FLUSH STRIDE=100
```

dat (plumed)

```
#! FIELDS time dist.x
0.000000  2.482382
1.000000 -2.960390
2.000000  3.263669
3.000000  2.972868
4.000000 -4.710670
5.000000 -1.715597
6.000000 -0.653605
7.000000  3.535680
8.000000  0.153147
```

DIST (plumed)

MUNI | CEITEC

Troubleshooting with Plumed

```
# Writes atom coordinates (including virtual atoms)
DUMPATOMS [...]
```
dat (plumed)

```
# Loads multiple molecules (or systems)
# By default, VMD appends trajectory
$ vmd -m file1.gro file2.gro
```
bash

MUNI | CEITEC

PLUMED Input File Format

- Calculate the distance between the peptide and box center
  - Only along the X-axis

```
# Indexes of all beads in the system are read from the ndx file
sys: GROUP NDX_FILE=index.ndx NDX_GROUP=System UNIQUE

# Creates a virtual atom in the center of the simulation box
c_sys: CENTER ATOMS=sys NOPBC

# Calculates the distance between the system center and an atom
dist: DISTANCE ATOMS=c_sys,11 COMPONENTS

PRINT ARG=dist.x FILE=DIST
FLUSH STRIDE=100
```

dat (plumed)

MUNI | CEITEC

Calculating the Distance

```
# Calculates the distance between the system center and an atom
dist: DISTANCE ATOMS=c_sys,11 COMPONENTS

PRINT ARG=dist.x FILE=DIST
FLUSH STRIDE=100
```

dat (plumed)

```
#! FIELDS time dist.x
0.000000 -0.996901
1.000000 -0.496520
2.000000 -0.572751
3.000000 -0.737869
4.000000 -1.210232
5.000000 -1.224067
6.000000 -1.403965
7.000000 -1.494100
8.000000 -1.582761
```

DIST (plumed)

MUNI | CEITEC

Calculating the Centroid

- Creates a virtual atom
  - CENTER can be used with arbitrary weights
  - Atom mass weighting can be used to calculate the center of mass (CoM)

```
# Creates virtual atoms (centroids of beads within each group)
c_sys: CENTER ATOMS=sys NOPBC
c_pep: CENTER ATOMS=Protein

# Writes a file with coordinates of selected beads
DUMPATOMS FILE=sys.gro ATOMS=c_sys,c_pep
```

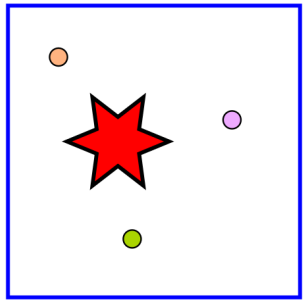
dat (plumed)

- Perodic boundary conditions (PBC) need to be treated correctly

MUNI | CEITEC

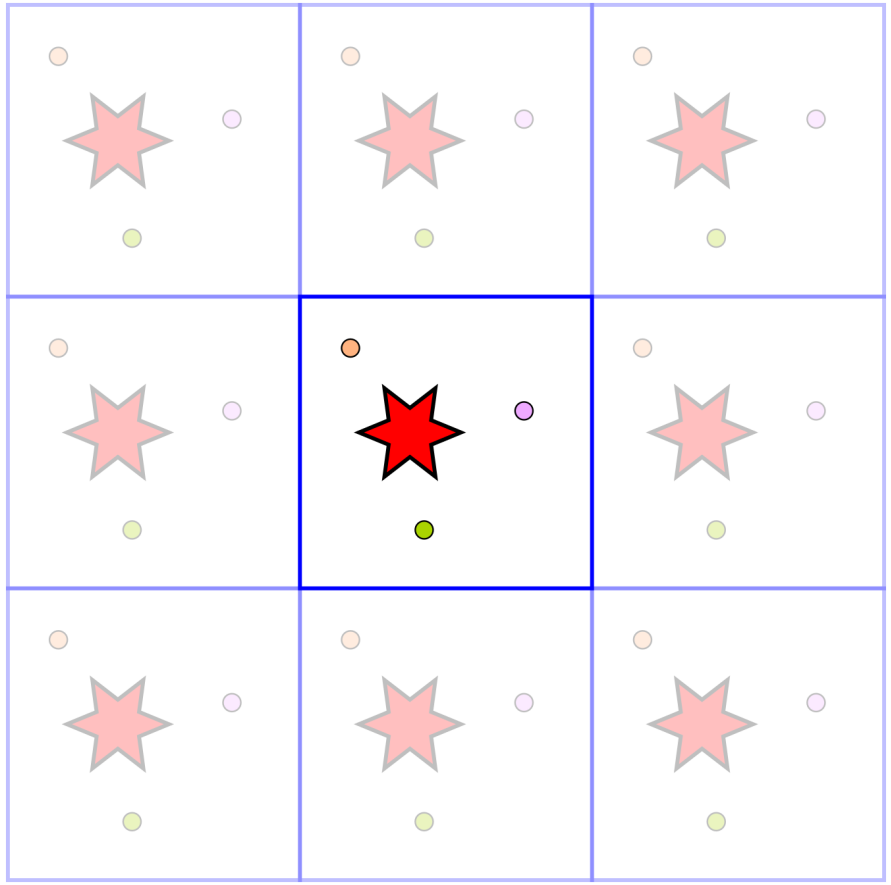
Calculating the Centroid

"Star" protein in a simulation box



Calculating the Centroid

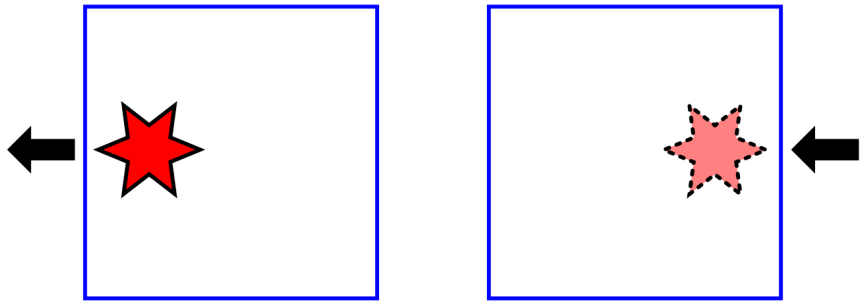
PBC replicate the original cell to mitigate finite size effects in simulations



MUNI | CEITEC

Calculating the Centroid

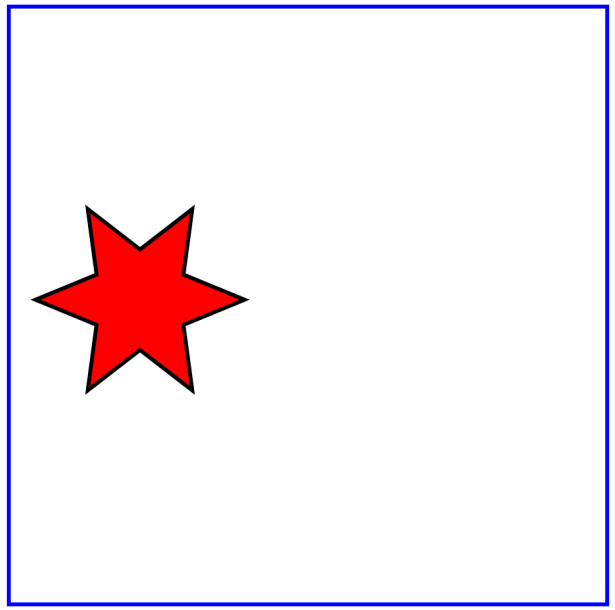
Atoms “reappear” on the opposite side when passing through boundaries



MUNI | CEITEC

Calculating the Centroid

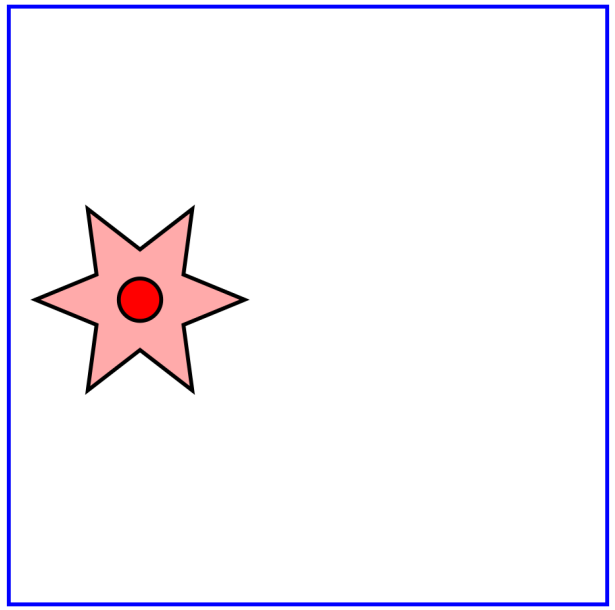
How to calculate the centroid of a molecule with PBC?



MUNI | CEITEC

Calculating the Centroid

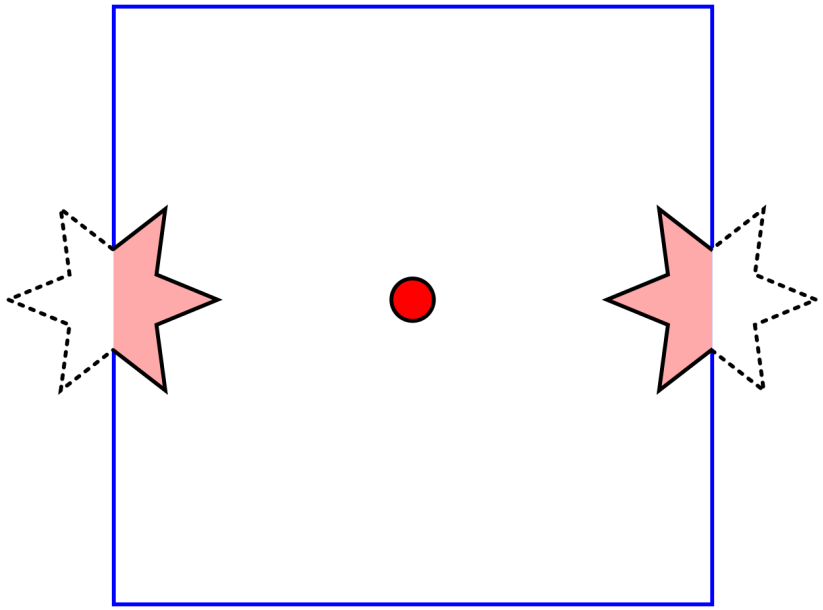
Mean position of all atoms



MUNI | CEITEC

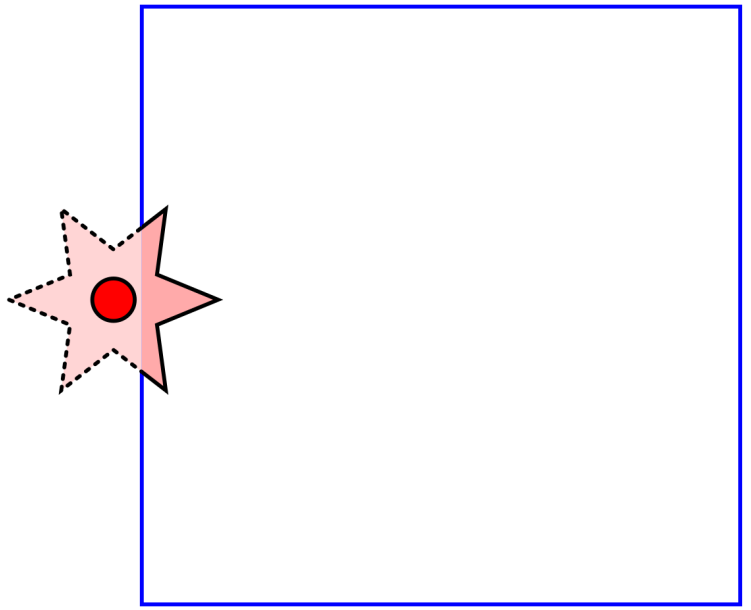
Calculating the Centroid

When a molecule is split across the boundary, the centroid will be outside the molecule



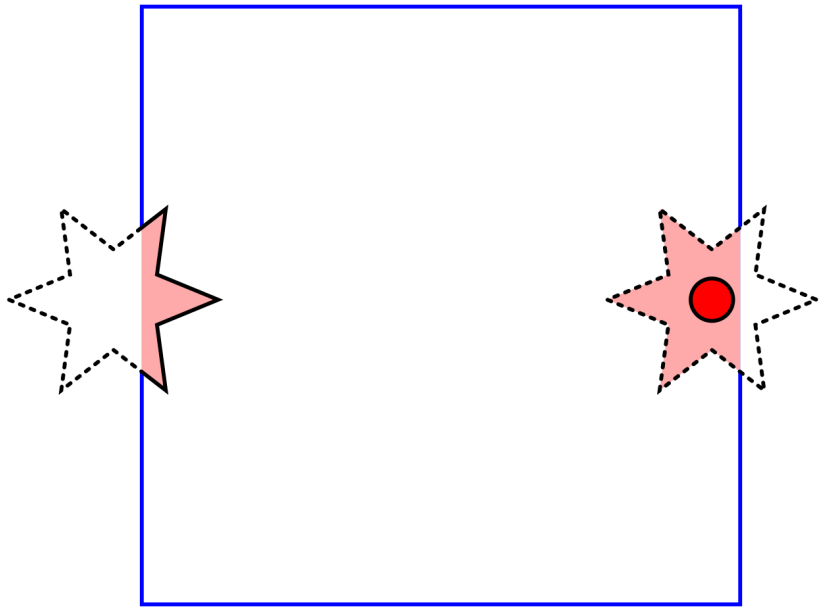
Calculating the Centroid

The solution is to rebuild the molecule before calculating the centroid …



Calculating the Centroid

… and then put the virtual atom back into the cell



Calculating the Centroid

- Is PBC treated automatically?
  - Does an action modify the coordinates?
  - NOPBC flag is available for many actions
- Actions to treat PBC
  - Executed in the order they are written in the PLUMED input file
  - WRAPAROUND
  - WHOLEMOLECULES

Calculating the Centroid

- WHOLEMOLECULES
  - Rebuilds molecules that are split across boundaries
  - Modifies the coordinates for subsequent actions
- WRAPAROUND
  - Rebuilds the system around selected atoms
  - Modifies the coordinates for subsequent actions
- DUMPATOMS
  - Action to write selected coordinates to a file
  - Can also write the coordinates of a virtual atom (for example, created by CENTER)

Calculating the Centroid

The radius of gyration usually makes sense when atoms used for the calculation are all part of the same molecule. When running with periodic boundary conditions, the atoms should b in the proper periodic image. This is done automatically since PLUMED 2.2, by considering the ordered list of atoms and rebuilding the broken entities using a procedure that is equivalent to that done in **WHOLEMOLECULES** . Notice that rebuilding is local to this action. This is different from **WHOLEMOLECULES** which actually modifies the coordinates stored in PLUMED.

In case you want to recover the old behavior you should use the NOPBC flag. In that case you need to take care that atoms are in the correct periodic image.

Calculating the Centroid

- PBC complicate the calculation of certain quantities
- Different actions handle PBC differently
- Different actions handle PBC differently depending on the version of the software

Calculating the Histogram

```
# Calculates the distance between the virtual atom and
dist: DISTANCE ATOMS=c_sys,c_pep COMPONENTS

# Calculate the histogram
HISTOGRAM ...
    LABEL=hist
    ARG=dist.x
    GRID_MIN=-4.5
    GRID_MAX=4.5
    GRID_BIN=100
    BANDWIDTH=0.1
...

# Write the calculated histogram to a file
DUMPGRID GRID=hist FILE=hist.dat
```

dat (plumed)

MUNI | CEITEC

Calculating the Free Energy

```
# Calculate the histogram
HISTOGRAM ...
    LABEL=hist
    ARG=dist.x
    GRID_MIN=-4.5
    GRID_MAX=4.5
    GRID_BIN=100
    BANDWIDTH=0.1
...

# Calculate the free energy (usually Gibbs)
fes: CONVERT_TO_FES GRID=hist TEMP=310

# Write the calculated free energy profile to a file
DUMPGRID GRID=fes FILE=fes.dat
```

$F(x) = -k_B T \ln H(x)$

dat (plumed)

MUNI | CEITEC

Convergence Tests

- Increase simulation length
  - Monitor if time-averaged quantities change
- Block averaging
  - Divide a trajectory into blocks of equal length and calculate averages for each block
  - Stratification / strata
- Multiple starting conditions
  - Simulations should yield similar averages
- Check for symmetries

Convergence Tests

```
# Simulation time or snapshot number
time: TIME

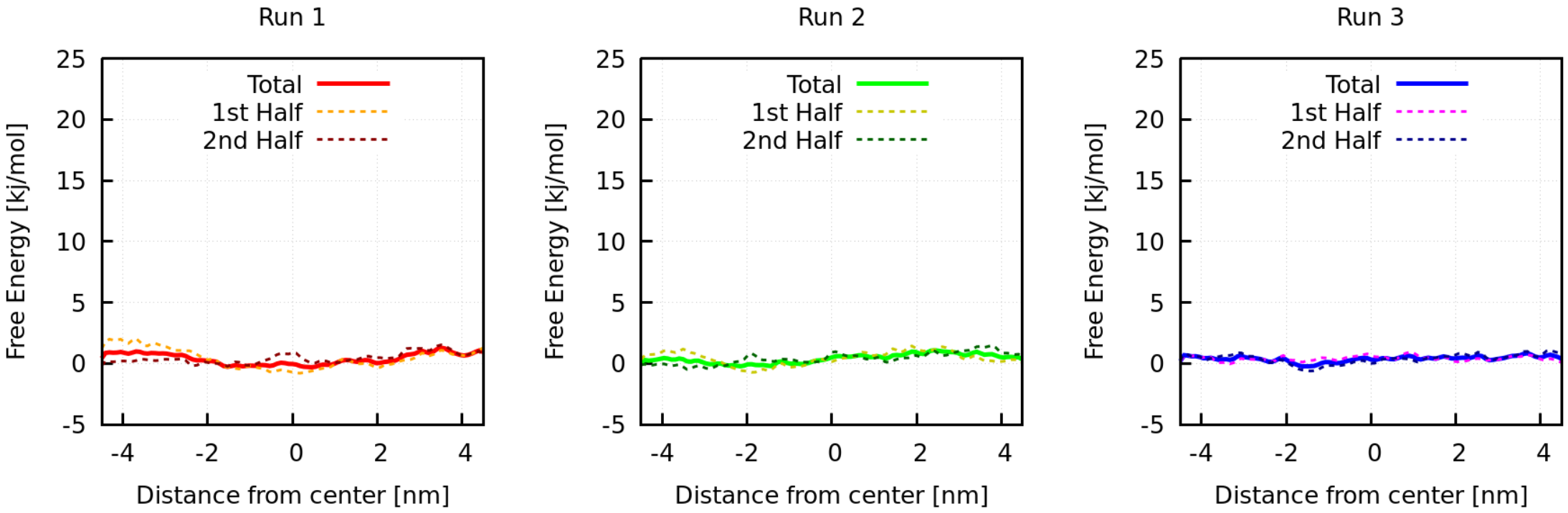
# First half of the simulation
UPDATE_IF ARG=time LESS_THAN=250
    hist_1: HISTOGRAM ARG=dist.x [...]
    fes_1: CONVERT_TO_FES GRID=hist_1 [...]
UPDATE_IF ARG=time END

# Second half of the simulation
UPDATE_IF ARG=time MORE_THAN=250
    hist_2: HISTOGRAM ARG=dist.x [...]
    fes_2: CONVERT_TO_FES GRID=hist_2 [...]
UPDATE_IF ARG=time END
```

dat (plumed)

MUNI | CEITEC

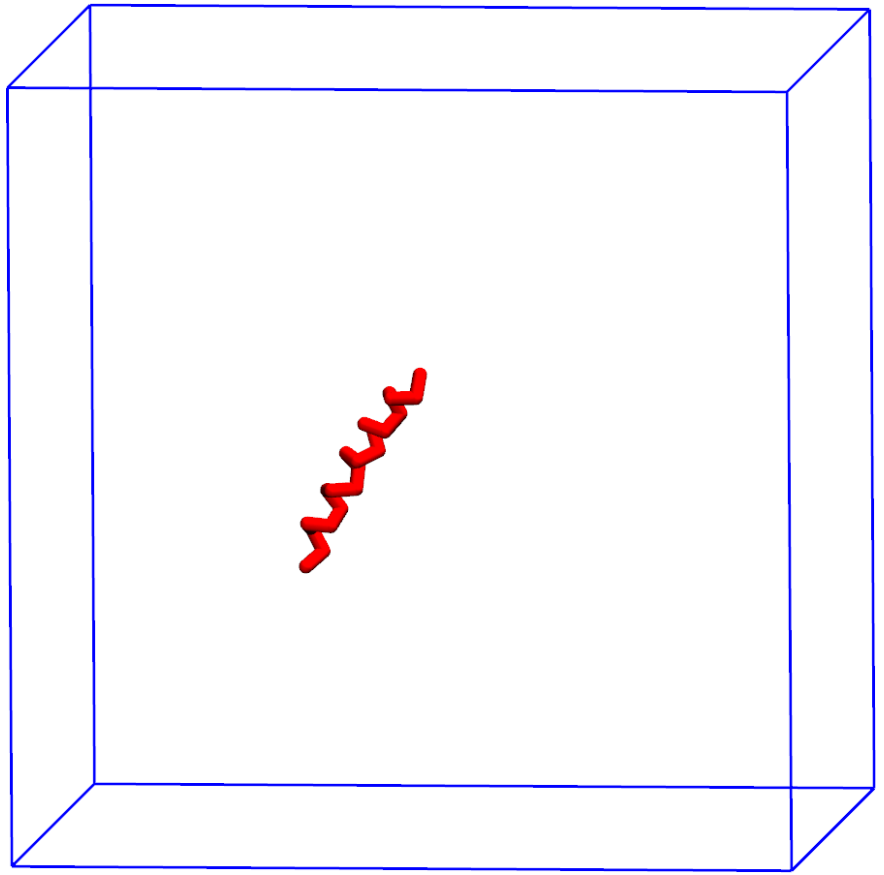
Free Energy and Convergence Test



MUNI | CEITEC

The Toy System

Calculating the orientation of the peptide



MUNI | CEITEC

Calculating the Angle in PLUMED

- Using MultiColvar:

```
# For convenience, MultiColvar can be used
angle: ZANGLES ATOMS1=nterm,cterm MEAN

# Printing the component
PRINT ARG=angle.mean FILE=ANGLE
```

dat (plumed)

- Alternatively, Colvar ANGLE can be used
  - Four atoms are used to define two vectors
  - FIXEDATOM is used to define the axis

Calculating the Angle in GROMACS

- Using pull code

```
pull                    = yes

pull_ngroups            = 2
pull_ncoords            = 1
pull_group1_name        = NTerm
pull_group2_name        = CTerm

pull_coord1_type        = umbrella
pull_coord1_geometry    = angle-axis
pull_coord1_groups      = 1 2
pull_coord1_vec         = 0 0 1
```

mdp (gromacs)

MUNI | CEITEC

Calculating the Angle in GROMACS

– A tpr file is needed for the calculation

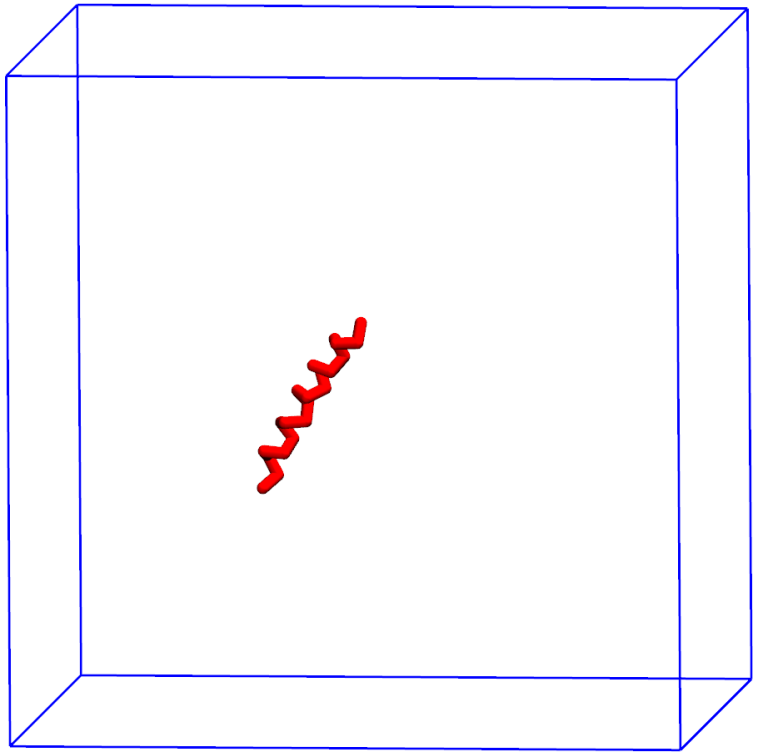
```bash
$ gmx grompp -f rerun.mdp \
             -n index.ndx \
             -p topol.top \
             -c equi.gro \
             -o rerun.tpr

$ gmx mdrun -deffnm rerun \
            -rerun prod.xtc
```

Task 2

Peptide in water

- Calculate the orientation of the peptide in the box



MUNI | CEITEC

Task 2

Peptide in water

- Calculate the orientation of the peptide in the box

```
pull                    = yes
pull_ngroups            = [...]
pull_ncoords            = [...]
pull_group1_name        = [...]
pull_group2_name        = [...]
pull_coord1_geometry    = [...]
pull_coord1_groups      = [...]
pull_coord1_vec         = [...]
```

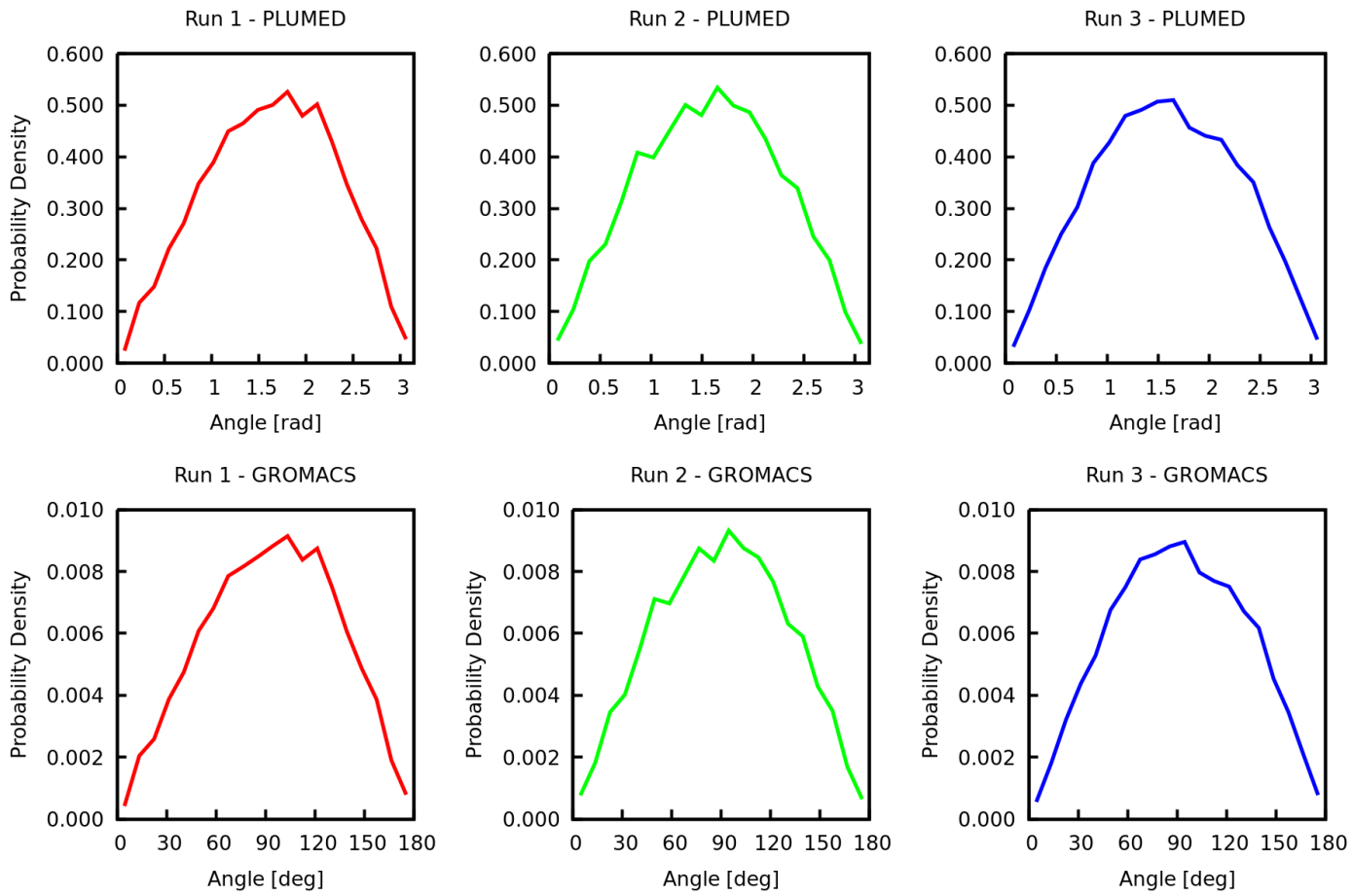
mdp (gromacs)

```
$ gmx grompp -f [...] -n [...] -p [...] -c [...] -o [...]
$ gmx mdrun -deffnm [...] -rerun [...]
```

bash

MUNI | CEITEC

Calculating the Angle



MUNI | CEITEC

Calculating the Cosine of an Angle

– Using CUSTOM function:

```
nterm: GROUP NDX_FILE=index.ndx NDX_GROUP=NTerm UNIQUE
cterm: GROUP NDX_FILE=index.ndx NDX_GROUP=CTerm UNIQUE

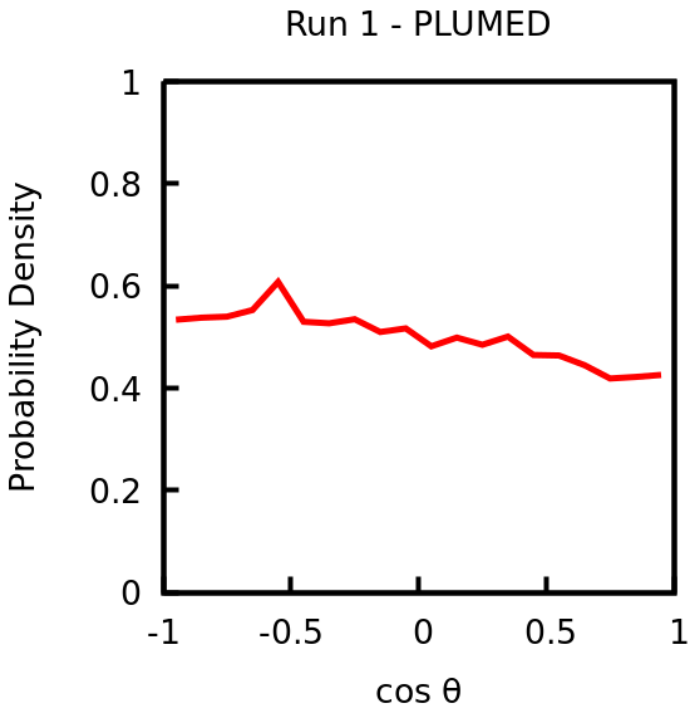
# Distance between N- and C- termini (peptide length)
dist1: DISTANCE ATOMS=nterm,cterm
# Components of the distance
dist2: DISTANCE ATOMS=nterm,cterm COMPONENTS

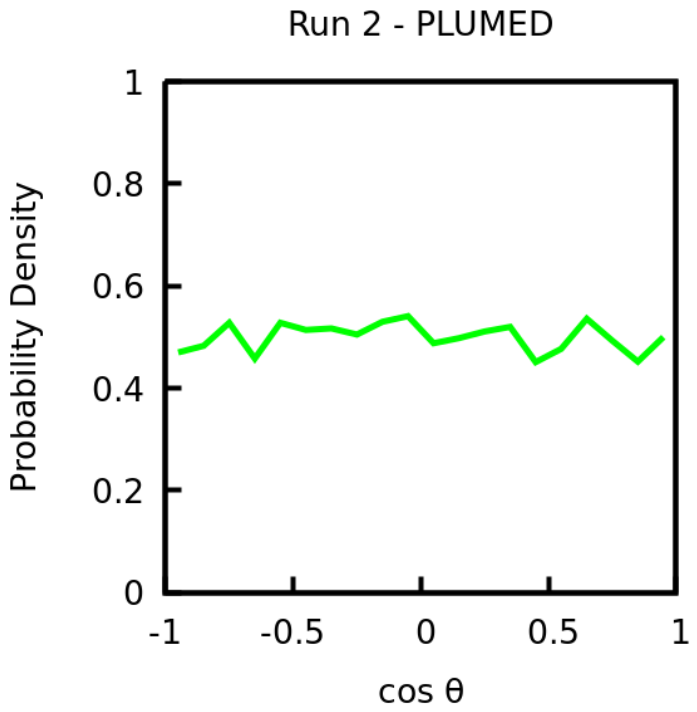
# Peptide orientation - cosine of the angle with respect to the Z-axis
cos_angle: CUSTOM ARG=dist1,dist2.z FUNC=y/x PERIODIC=NO
```

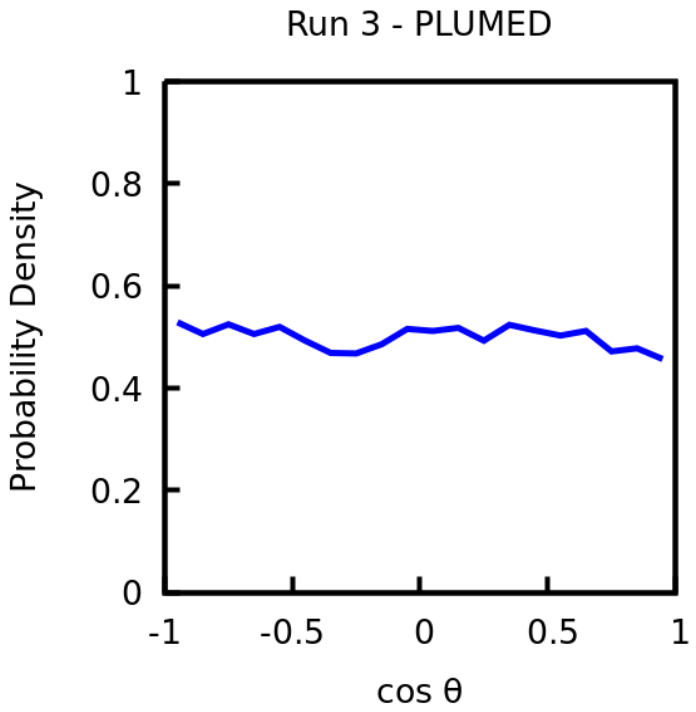
dat (plumed)

MUNI | CEITEC

Calculating the Cosine of an Angle







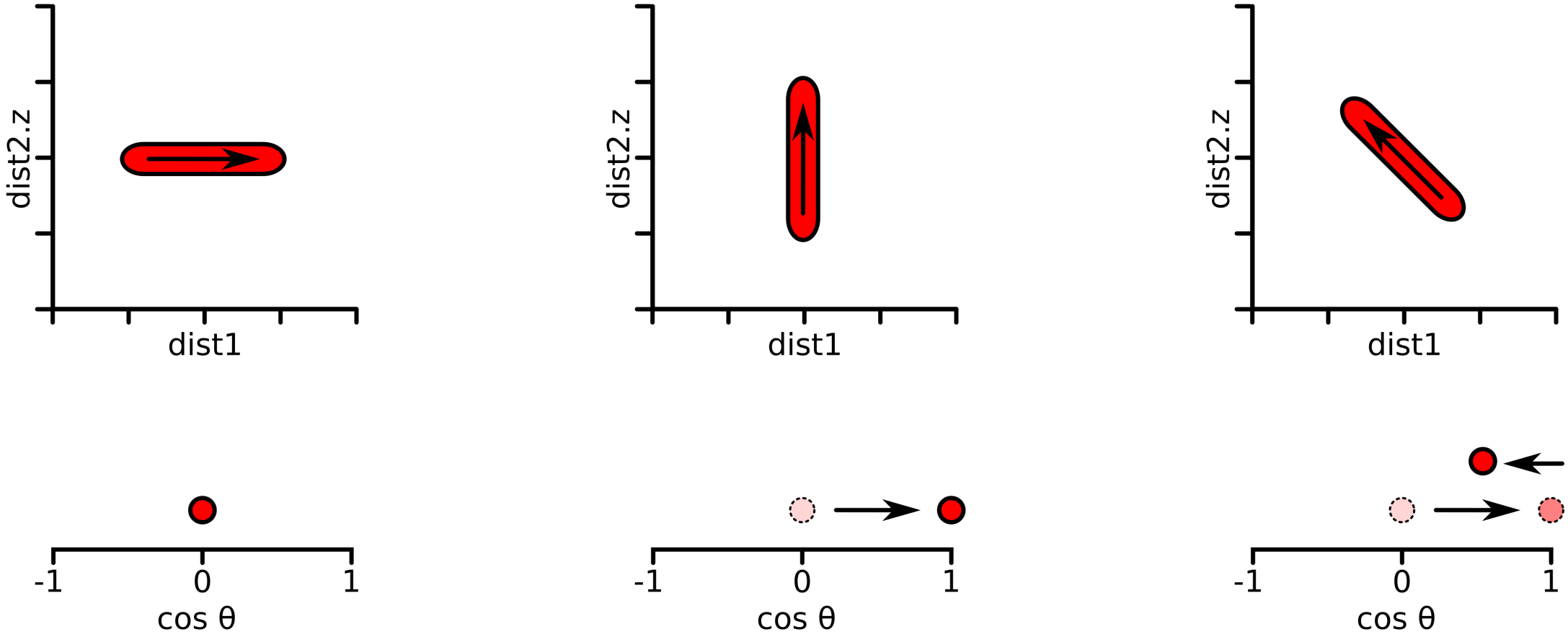
MUNI | CEITEC

Collective Variable (CV)

- A parameter that describes the progress of a process
  - Reduces the complexity of many-particle systems to a few key degrees of freedom
  - Must accurately represent the process or transition of interest
- CV must be differentiable, smooth, and continuous
- Examples of common collective variables:
  - Distance between atoms/beads or molecules
  - Angle or dihedral angle
  - Coordination number
  - Radius of gyration

Calculating the Cosine of an Angle

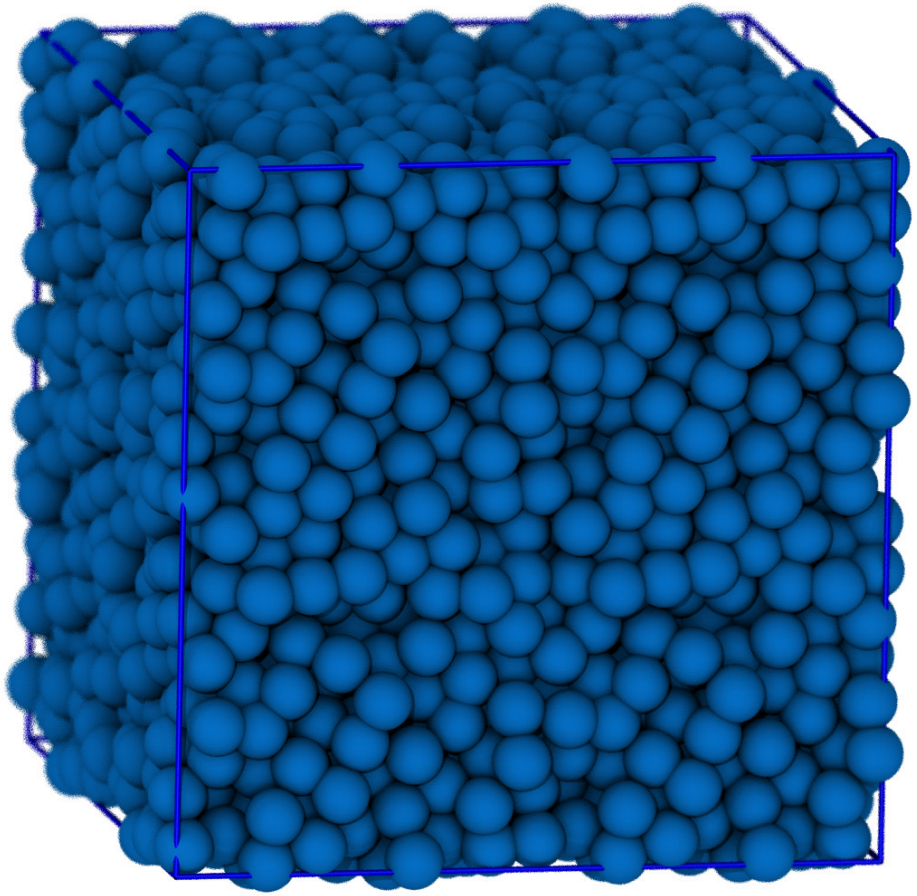
Collective variable must be continuous



MUNI | CEITEC

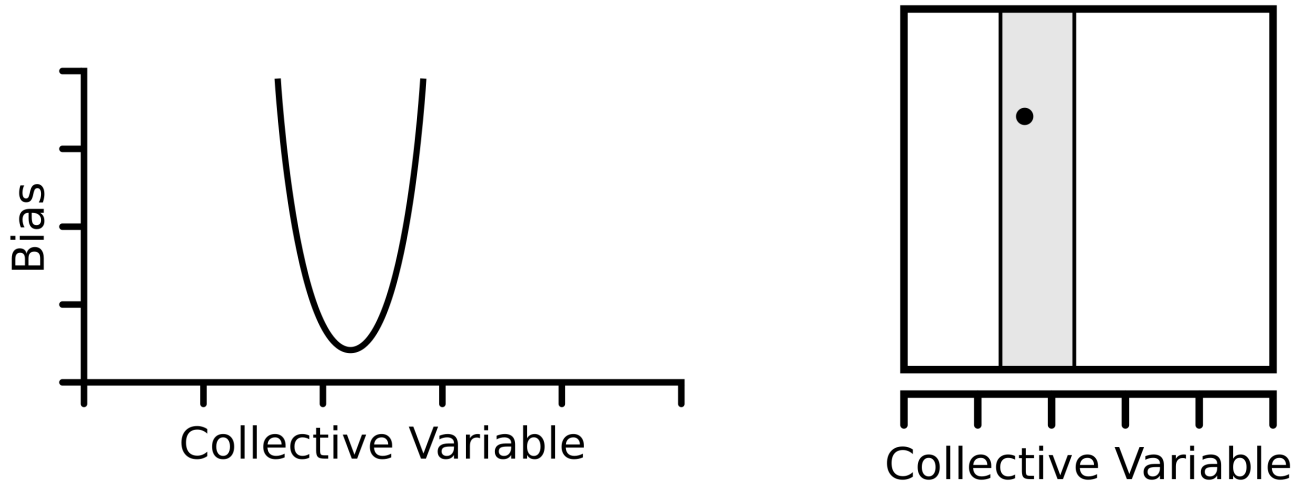
The Toy System

Water Box



MUNI | CEITEC

Harmonic Restraint in GROMACS

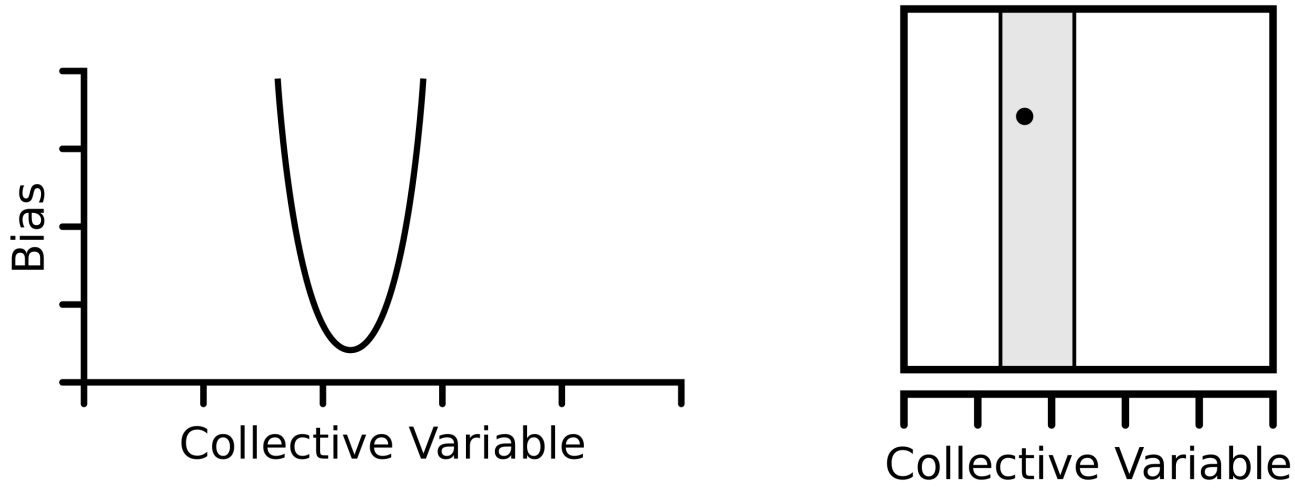



```
pull_coord1_type        = umbrella
pull_coord1_rate        = 0.0
pull_coord1_k           = 8000
pull_coord1_start       = no
pull_coord1_init        = 0.0
```

mdp (gromacs)

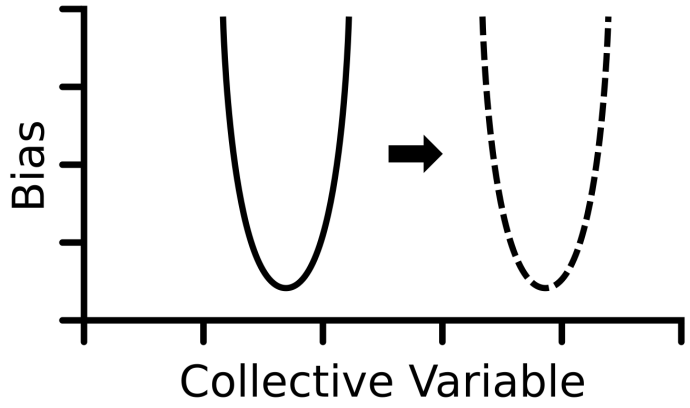
MUNI | CEITEC

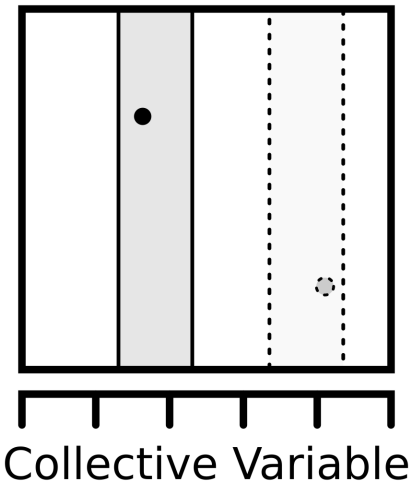
Harmonic Restraint in PLUMED




```
res: RESTRAINT ARG=dist.z AT=0.0 KAPPA=8000
```
dat (plumed)

Moving Harmonic Restraint in GROMACS



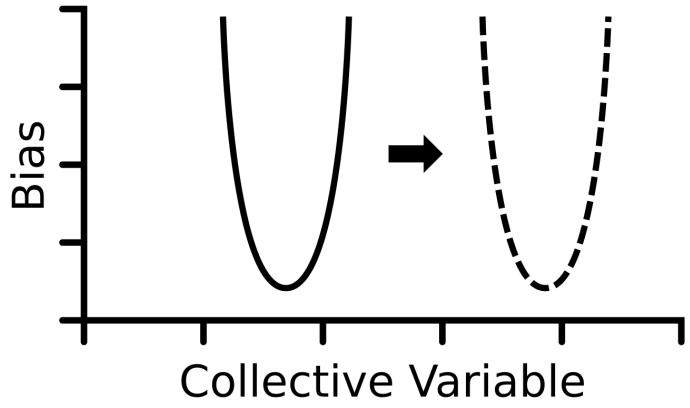


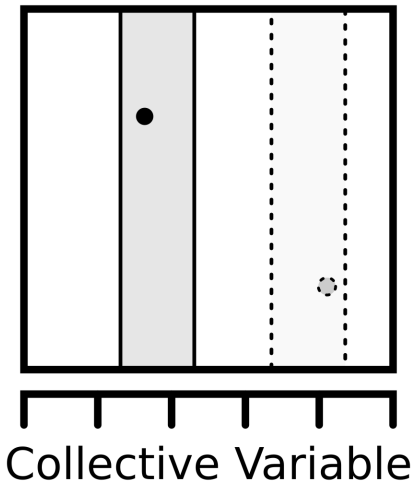

```
pull_coord1_type.        = umbrella
pull_coord1_rate         = -0.000034
pull_coord1_k            = 8000
pull_coord1_start        = no
pull_coord1_init         = 0.0
```

mdp (gromacs)

MUNI | CEITEC

Moving Harmonic Restraint in PLUMED



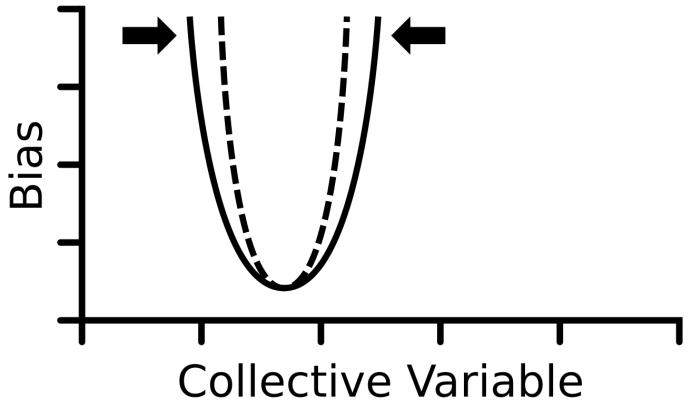


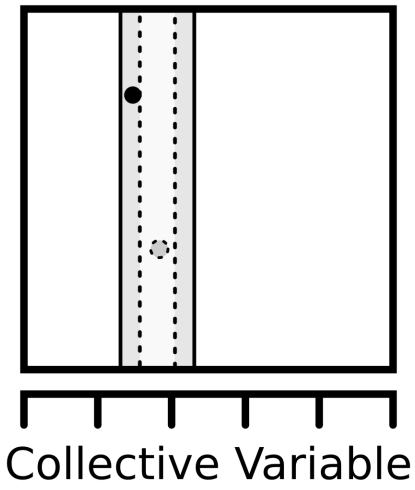

```
MOVINGRESTRAINT ...
    LABEL=res
    ARG=dist.z
    STEP0=0         AT0=-2.5   KAPPA0=8000
    STEP1=5000000   AT1=2.5    KAPPA1=8000
...
```

dat (plumed)

MUNI | CEITEC

Moving Restraints in PLUMED



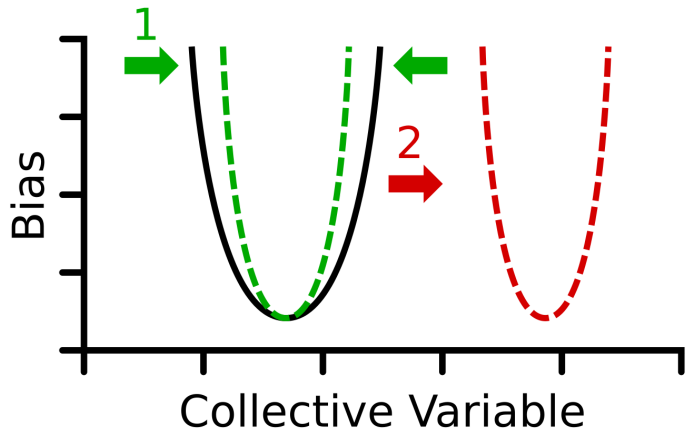


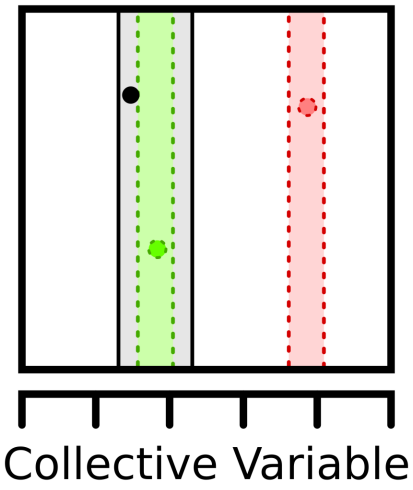

```
MOVINGRESTRAINT ...
    LABEL=res
    ARG=dist.z
    STEP0=0          AT0=-2.5   KAPPA0=2500
    STEP1=5000000    AT1=-2.5   KAPPA1=8000
...
```

dat (plumed)

MUNI | CEITEC

Moving Restraints in PLUMED



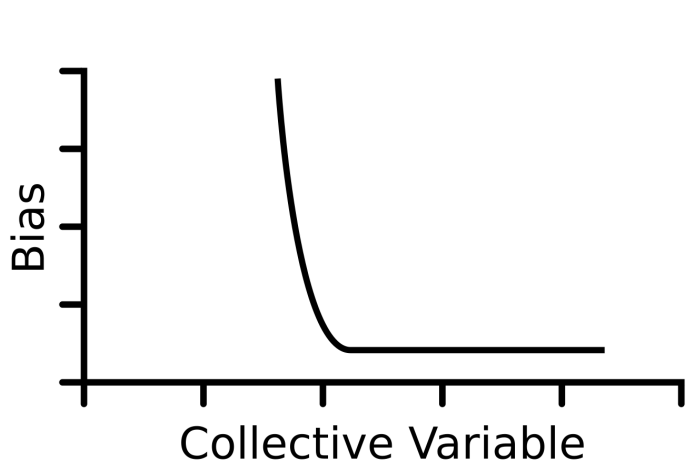


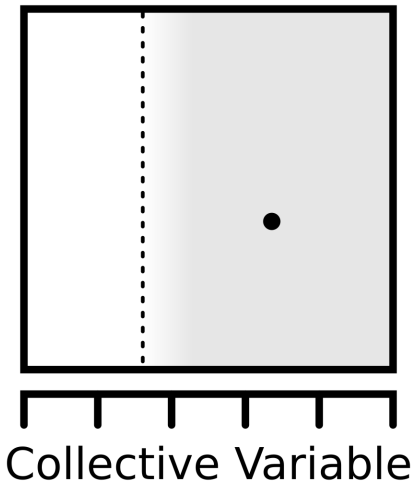

```
MOVINGRESTRAINT ...
    LABEL=res
    ARG=dist.z
    STEP0=0          AT0=-2.5   KAPPA0=2500
    STEP1=10000      AT1=-2.5   KAPPA1=8000
    STEP2=5000000    AT2=2.5    KAPPA2=8000
...
```

dat (plumed)

MUNI | CEITEC

Flat-bottom Restraints in GROMACS



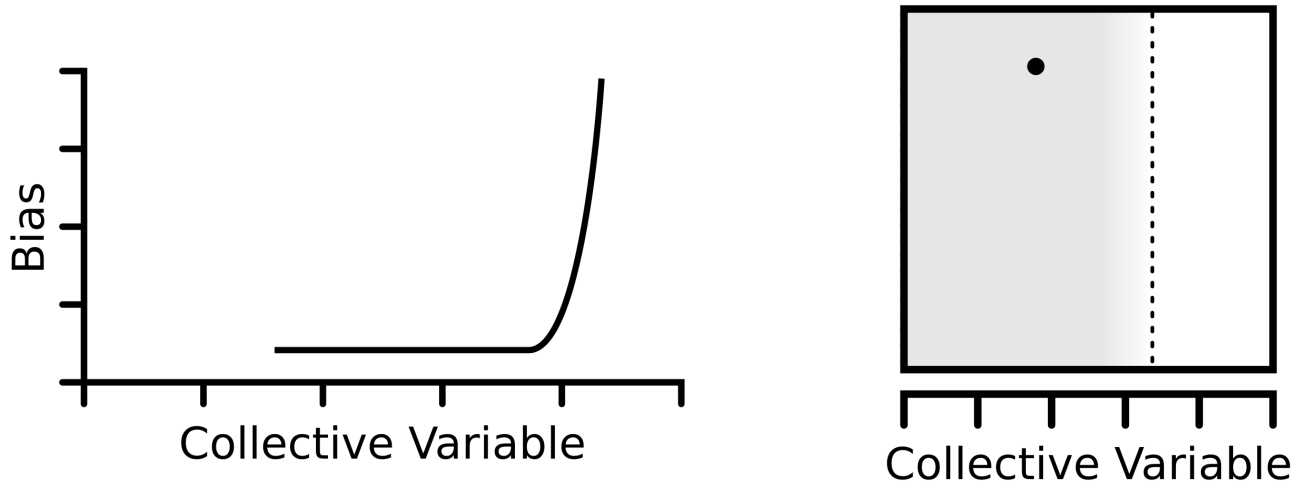



```
pull_coord1_type        = flat-bottom-high
pull_coord1_rate        = 0.0
pull_coord1_k           = 8000
pull_coord1_start       = no
pull_coord1_init        = 0.0
```

mdp (gromacs)

MUNI | CEITEC

Flat-bottom Restraints in GROMACS

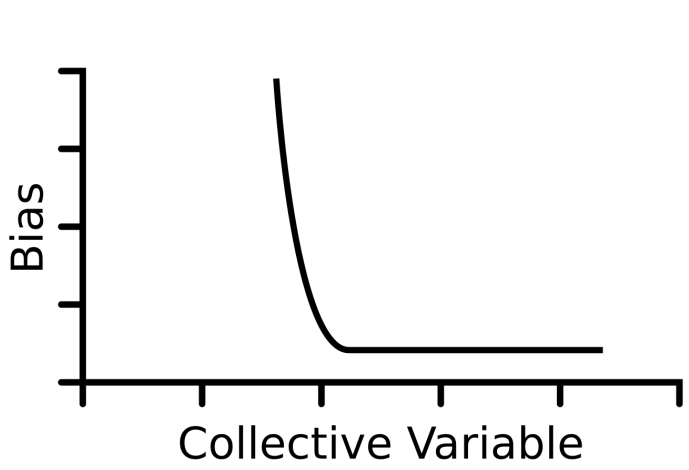


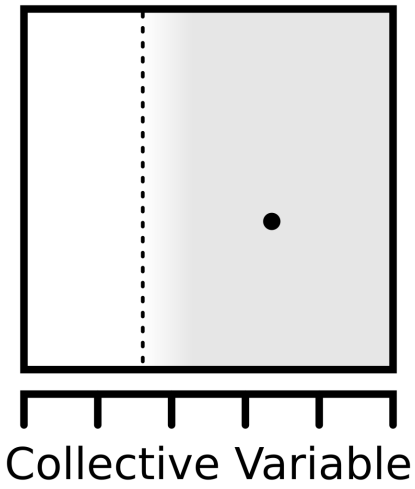

```
pull_coord1_type        = flat-bottom
pull_coord1_rate        = 0.0
pull_coord1_k           = 8000
pull_coord1_start       = no
pull_coord1_init        = 0.0
```

mdp (gromacs)

MUNI | CEITEC

Flat-bottom Restraints in PLUMED

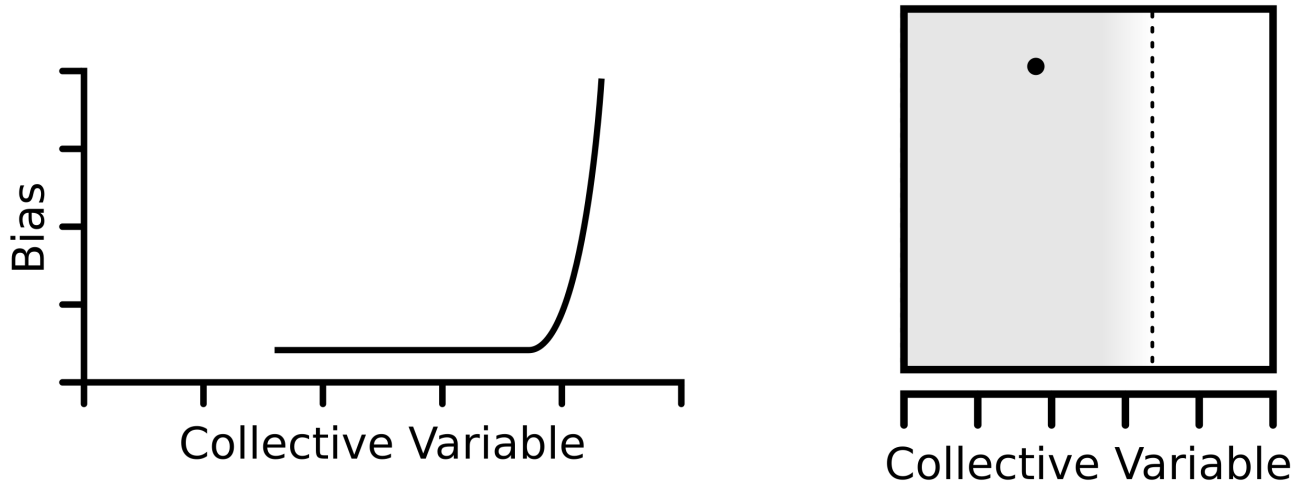





```
res: LOWER_WALLS ARG=dist.z AT=0.0 KAPPA=8000
```
dat (plumed)

MUNI | CEITEC

Flat-bottom Restraints in PLUMED

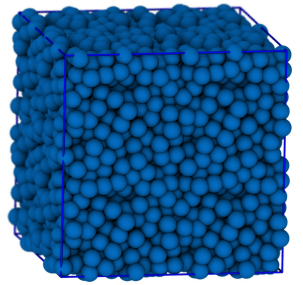


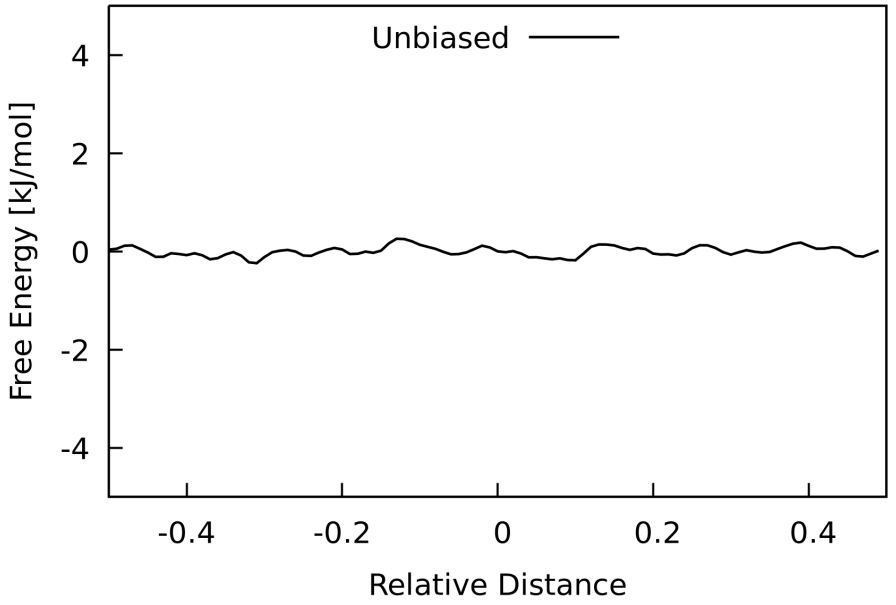

```
res: UPPER_WALLS ARG=dist.z AT=0.0 KAPPA=8000
```
dat (plumed)

MUNI | CEITEC

Free Energy from Unbiased Simulation

Water Translation





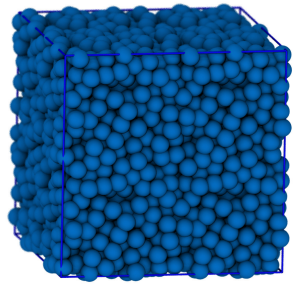

```
dist: DISTANCE ATOMS=c_sys,1 SCALED_COMPONENTS
HISTOGRAM […]
fes: CONVERT_TO_FES […]
DUMPGRID GRID=fes FILE=fes.dat
```

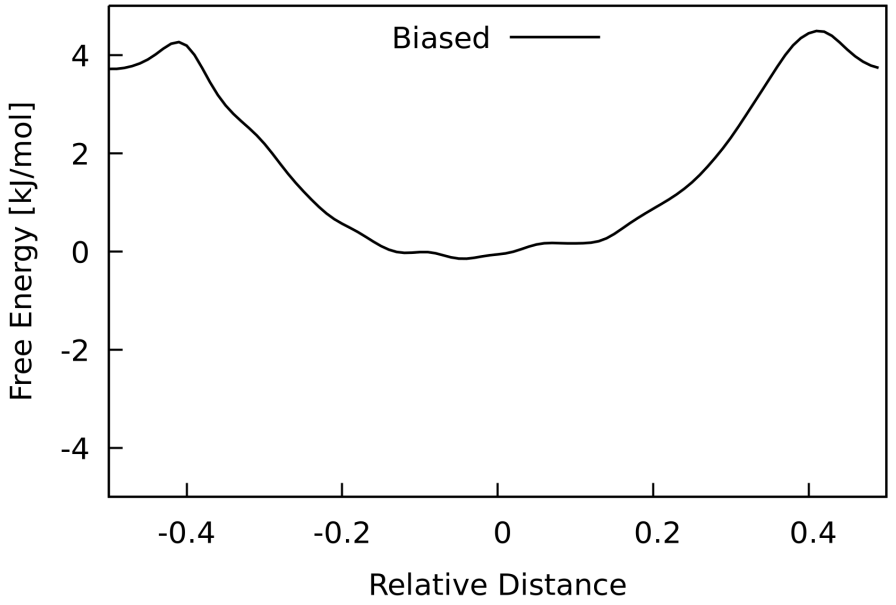
dat (plumed)

MUNI | CEITEC

Free Energy from Biased Simulation

Water Translation





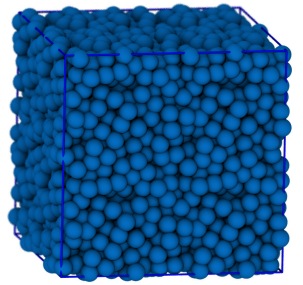

```
lwall: LOWER_WALLS ARG=dist.c AT=-0.1 KAPPA=50
uwall: UPPER_WALLS ARG=dist.c AT=0.1  KAPPA=50
```

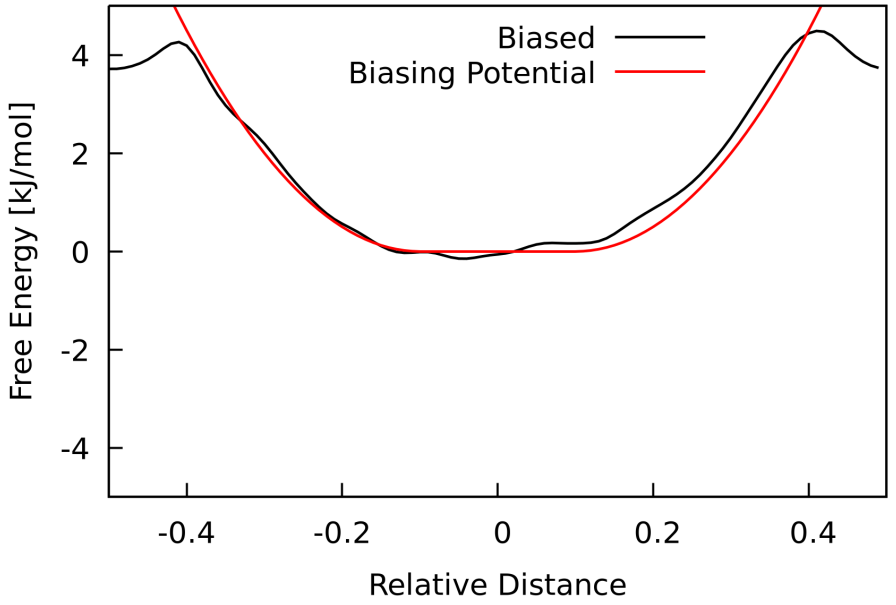
dat (plumed)

MUNI | CEITEC

Free Energy from Biased Simulation

Water Translation





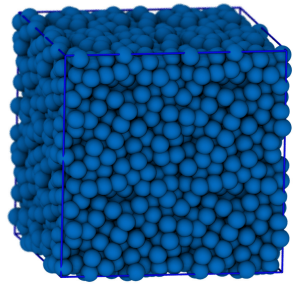

```
lwall: LOWER_WALLS ARG=dist.c AT=-0.1 KAPPA=50
uwall: UPPER_WALLS ARG=dist.c AT=0.1  KAPPA=50
```

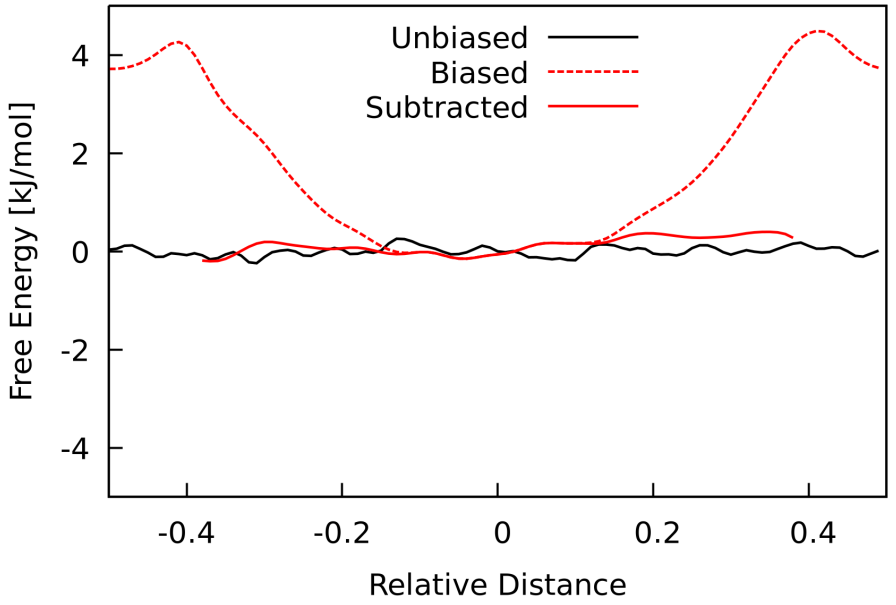
dat (plumed)

MUNI | CEITEC

Free Energy from Biased Simulation

Water Translation



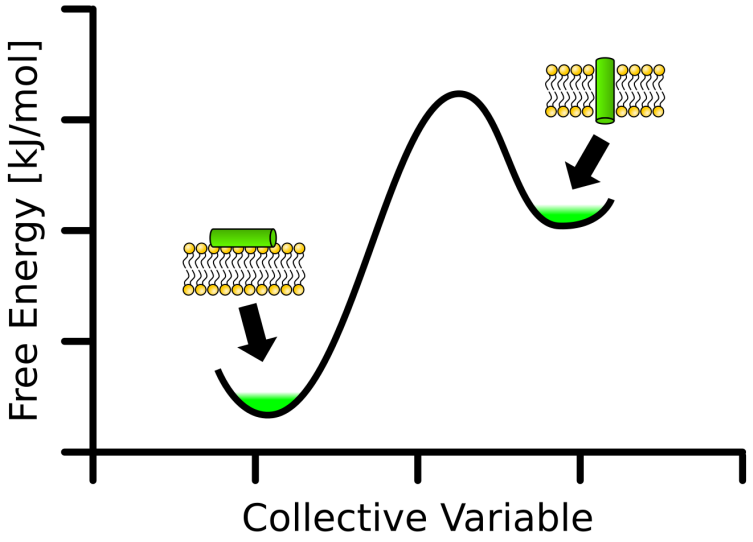



```
lwall: LOWER_WALLS ARG=dist.c AT=-0.1 KAPPA=50
uwall: UPPER_WALLS ARG=dist.c AT=0.1  KAPPA=50
```

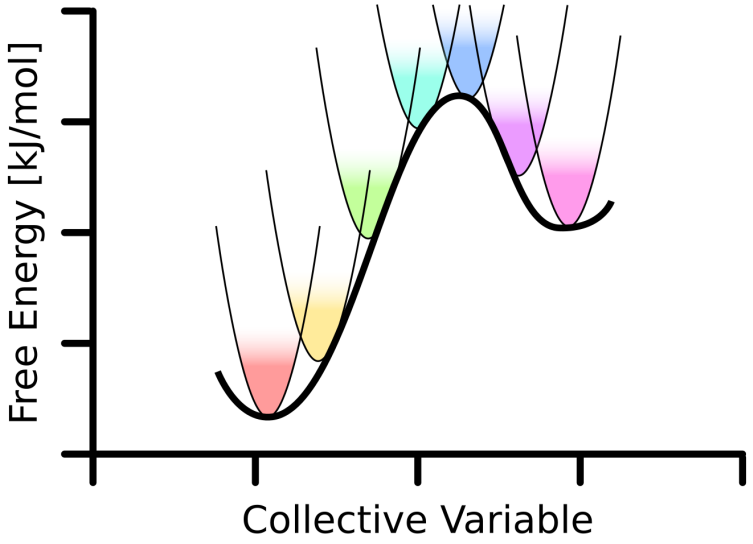
dat (plumed)

MUNI | CEITEC

Umbrella Sampling

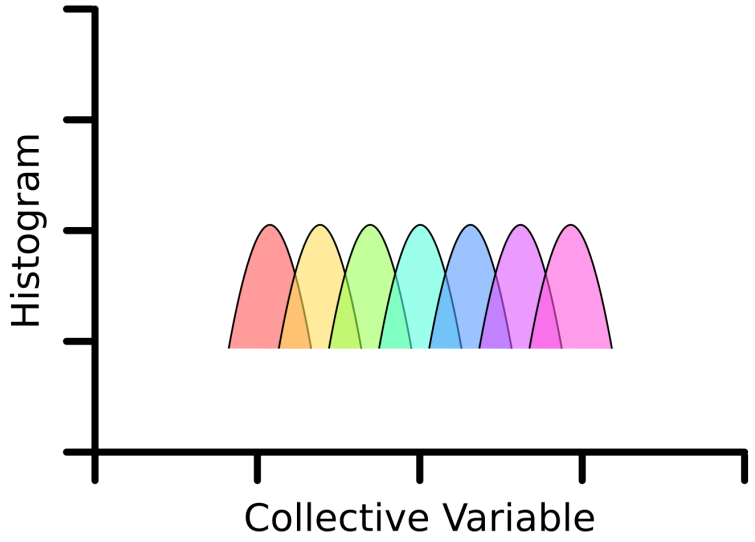


Umbrella Sampling



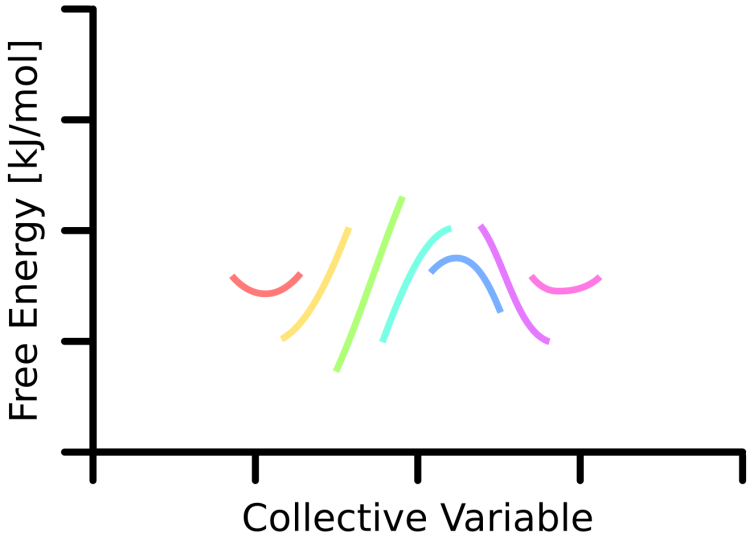
MUNI | CEITEC

Umbrella Sampling

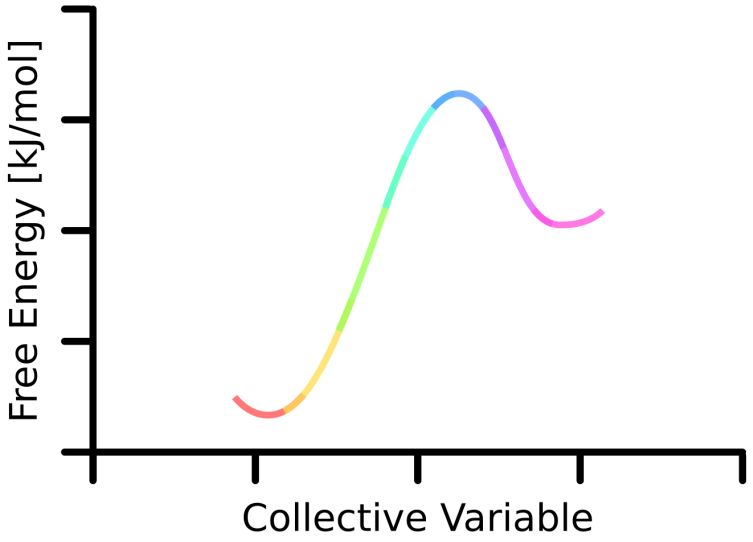


MUNI | CEITEC

Umbrella Sampling

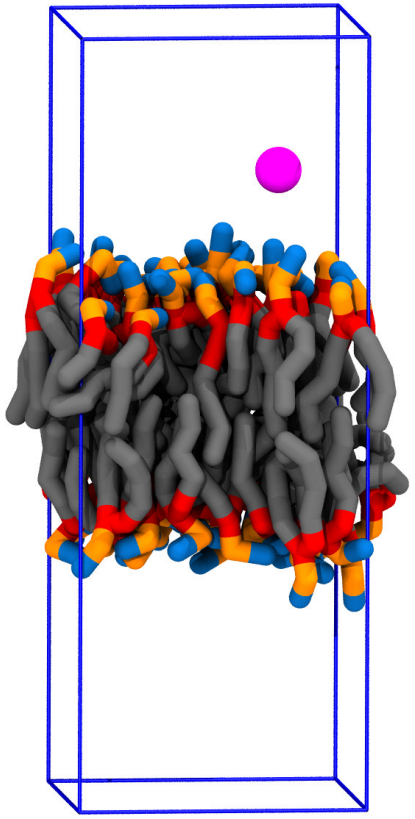


Umbrella Sampling



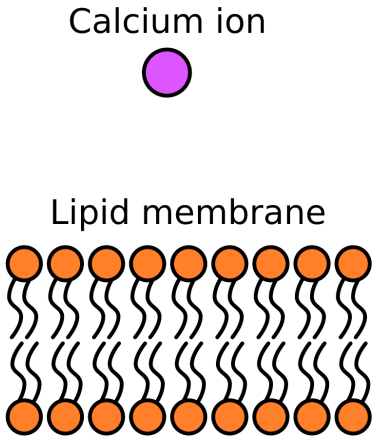
The Toy System

Calculating ion permeation across a lipid bilayer

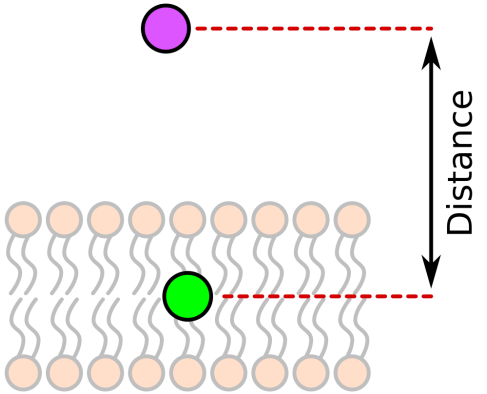


MUNI | CEITEC

Ion Permeation

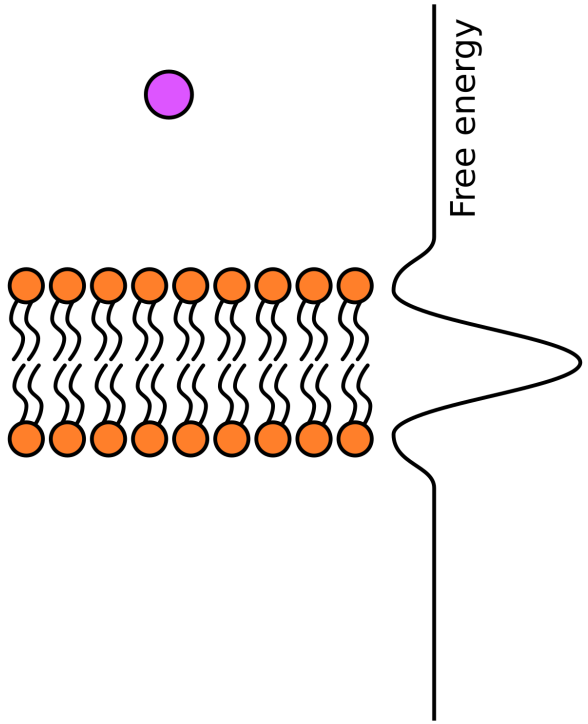


Ion Permeation



MUNI | CEITEC

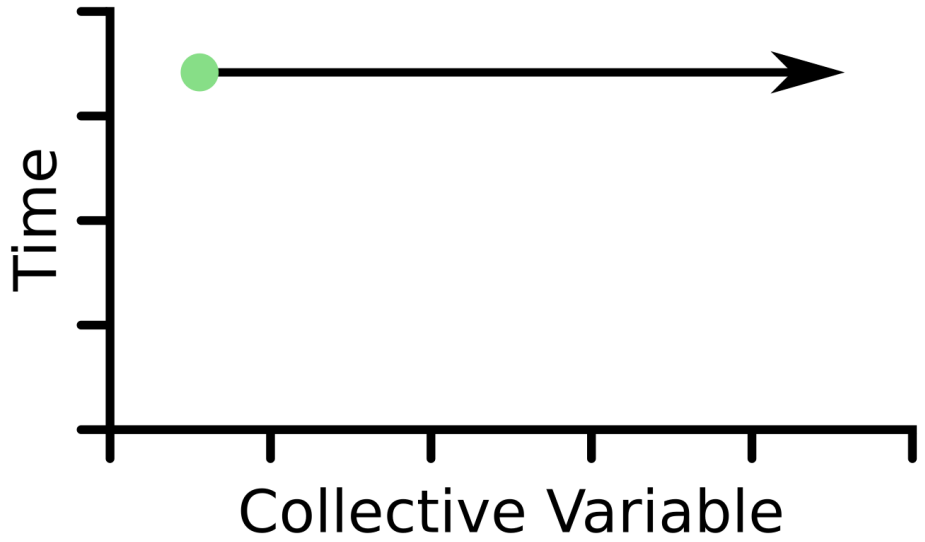
Ion Permeation



MUNI | CEITEC

Umbrella Sampling

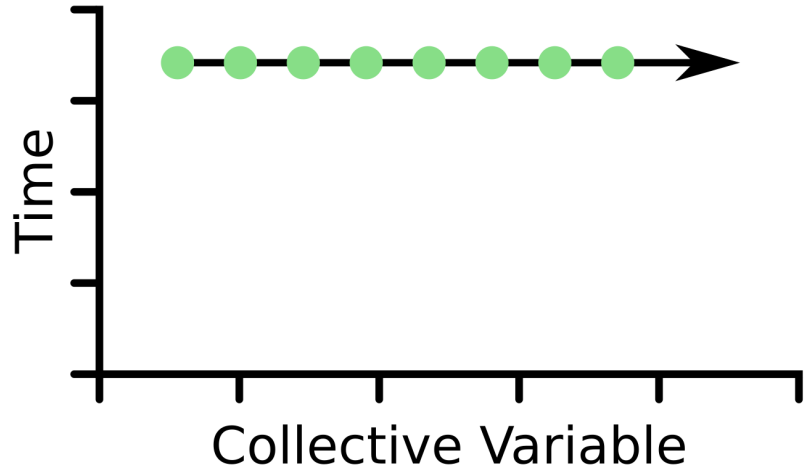
From the initial configuration, the system is “pulled” along the collective variable



“Pulling” is usually done at a constant rate, typically with a harmonic restraint

Umbrella Sampling

Simulation snapshots are extracted from the trajectory



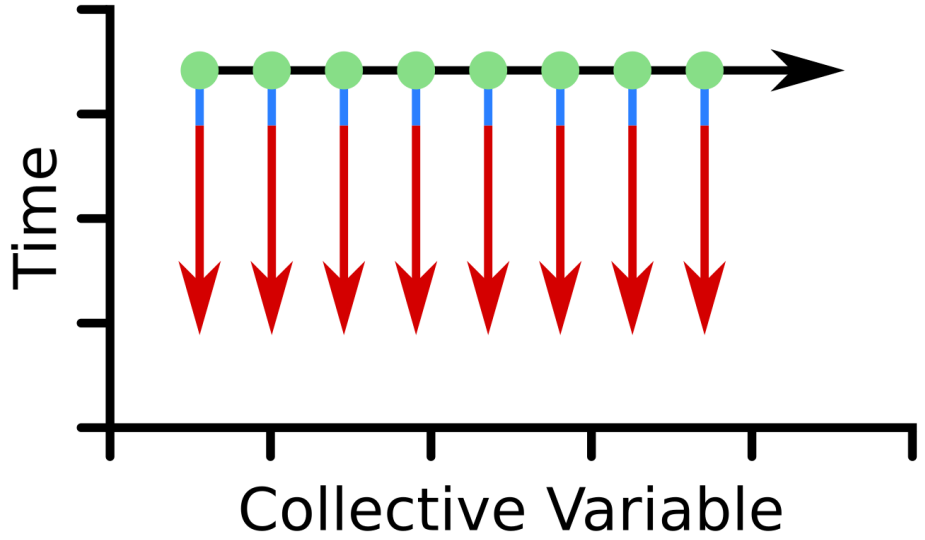

```
nstxout                 = 100000 ; frequency of writing coordinates (trr)
nstvout                 = 100000 ; frequency of writing velocities (trr)
pull-nstxout            = 100000 ; frequency of writing the CV value
pull_coord1_type        = umbrella ; harmonic restraint
pull_coord1_rate        = -0.000034
```

mdp (gromacs)

MUNI | CEITEC

Umbrella Sampling

New simulations are started from all the generated configurations

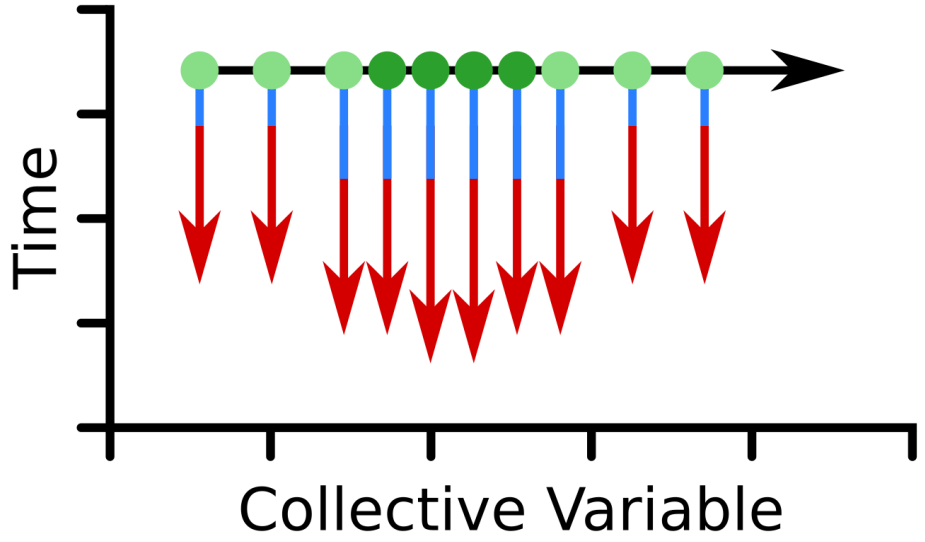


Each simulation (window) typically has an equilibration part (blue)
and a production part (red) that is used for data collection

MUNI | CEITEC

Umbrella Sampling

The spacing, equilibration time, and force constants can be optimized



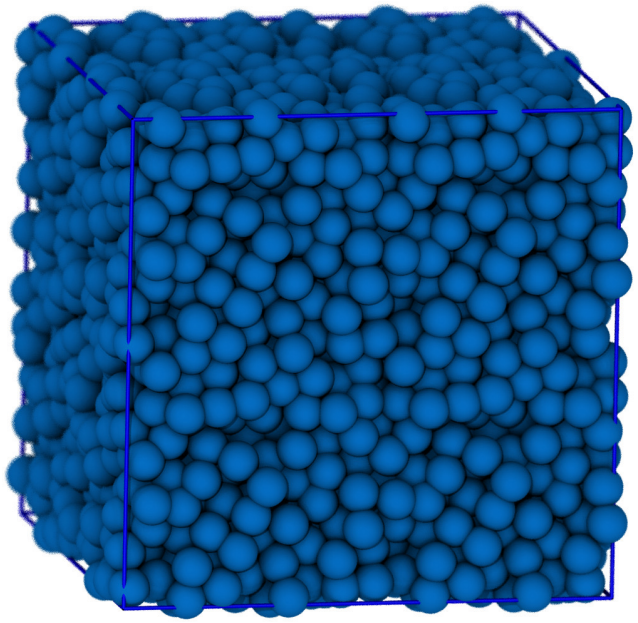
The distribution of windows does not have to be uniform
Each window can also have a different force constant
The simulation duration for each window can also be different

MUNI | CEITEC

Task 3

Water Box

- Select a bead and put in the box center
- Restrain the bead
  - Within a plane
  - Within a slab
  - Along an axis
  - Within a cylinder



```bash
$ gmx editconf -f [...] -c -center [...]
$ gmx trjconv -s [...] -f [...] -center -pbc atom
```


MUNI | CEITEC

Task 4

Protein in Water

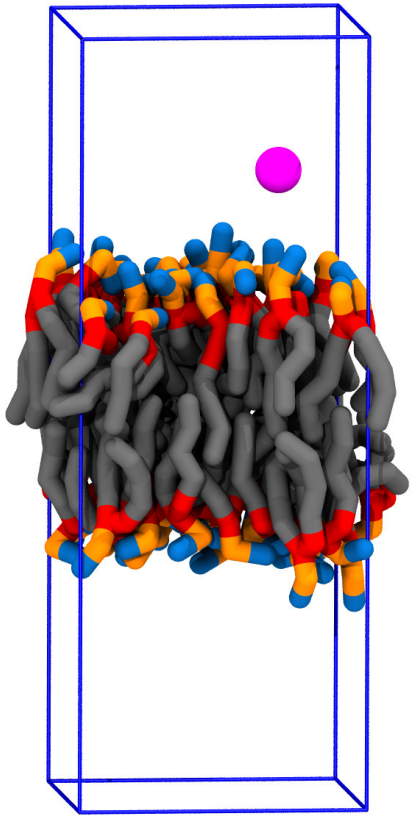
- Apply a restraint on the distance between two residues
  - Open a simulation snapshot in VMD
  - Pick two atoms and measure the distance between them
  - Use PLUMED to apply a restraint on this distance

MUNI | CEITEC

Task 5

Lipid bilayer

- Adsorption of calcium ion on a lipid bilayer
- Perform an unbiased MD simulation
- Calculate the free energy of adsorption

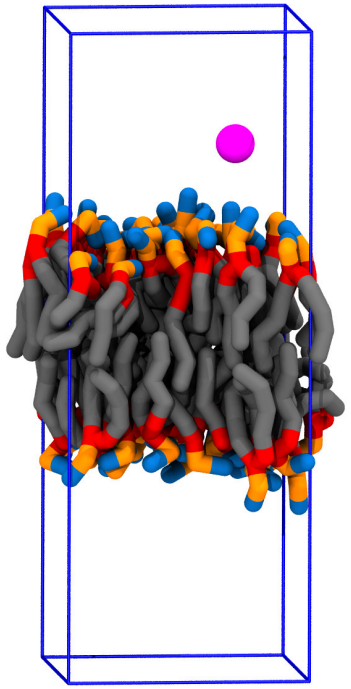


MUNI | CEITEC

Task 6

Lipid bilayer

- Perform a pulling simulation using GROMACS and PLUMED
  - Pull an ion across the membrane (in Z-axis)
  - Use pull code (GROMACS) or MOVINGRESTRAINT (PLUMED)



Print the trajectory snapshots with the same frequency as the collective variable

MUNI | CEITEC